

CIE

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TITLE: Real-Time Collaborative Development Environment

SCHOOL: Chitkara University, Rajpura, Punjab

**DETAIL OF THE STUDENT/MENTOR WHO UPLOADED THE COPYRIGHT ON GOVT PORTAL NEED TO
SHARE LOGIN AND PASSWORD WITH MENTORS**

Name: Aayush Sharma

Roll No/ Employee Code: 2210991138

ALL MEMBERS DETAILS INCLUDING SUPERVISOR

NAME	EMP CODE/ROLL	ADDRESS WITH EMAIL AND MOBILE NO.
Yashasvi Nag	2210990975	yashasvi975.be22@chitkara.edu.in , 7009770580
Vrinda Verma	2210990969	vrinda969.be22@chitkara.edu.in ,9991200885
Jashanjit Kaur	2210990440	jashanjit440.be22@chitkara.edu.in , 7889198482
Aayush Sharma	2210991138	aayush1138.be22@chitkara.edu.in ,9729248257
Dr. Shikha Tuteja		shikha.1290@chitkara.edu.in , 9915369888

ABSTRACT

1.1 Problem Background

Concern for mental health is fully realised; yet, still many receive no care and treatment or have to go through suffering unable to get support

Also, concern about mental health affects many people today. Plenty of people do not seek therapy because they fear being judged or do not have proper access to professional care.

million or more people in India do not seek help for these reasons.

Traditional therapy is also time-consuming and expensive for clients.

All of these make it imperative to create a system that is readily accessible, private, and can respond quickly to a person's needs through the use of technology or other methods.

By using digital platforms or artificial intelligence systems to assist with the expression of feelings and providing immediate support to users, the company aims to provide a channel for individuals to communicate with someone about what they are feeling and then use that channel again later, creating an early step toward improving one's mental health or, at least, being willing to seek assistance when needed.

1.2 Objective of the Project

The primary objective of the project is to develop a mental health support system that interacts with users in real-time and capable of delivering customized responses. It accomplishes this by integrating chatbot capabilities with emotion detection through facial recognition and continual tracking of the user's emotional state. Ultimately, this will allow the emotional support system to better understand the emotional state of the user while offering tools and suggestions in a supportive and user-friendly manner.

1.3 Proposed Solution / System Overview

MERN stack is a full-stack framework for building web applications. The system will provide a chat interface for users to share ideas and receive information from the chatbot based on the information they provide through the chat interface.

The system will have a facial recognition feature as well. Users will be able to upload pictures of themselves to determine their face and also use the camera on their device for emotion tracking.

The information from all interactions will be stored securely online and used to create reports and track emotion patterns. In addition, the system will have a user-friendly interface that will provide easy access to resources for all users.

1.4 Key Features or Functionality

- **Chat support system:** Users will have a real-time conversation with an automated chat support system and receive relevant support based on the information that they provide through the chat interface.
- **Document history:** All chat histories will be saved so that users may track their emotional changes over time to better understand their emotions.
- **Object recognition:** Users will be identified by uploading a picture or using the camera on the device to detect emotion based on face.
- **Generate reports:** Generate reports and summaries of user interactions with emotions and the effect on the wellness of the user.

- **List of wellness suggestions:** The system will also provide users with easy-to-use and tailored suggestions to help them manage their stress and/or emotions.
- **Evidence-Based Mental Health Assessments:** Clinically-grounded self-assessment tests based on validated psychological instruments
- **Severity-Coded Results** – Color-coded severity levels for quick interpretation:
 - Mild
 - Moderate
 - Severe
 - Extreme (with clinical alerts)
 - Critical Alerts – High-risk indicators with recommended next steps



Figure 1 –Shows the main interface of the system with access to chat, profile, and analysis features.

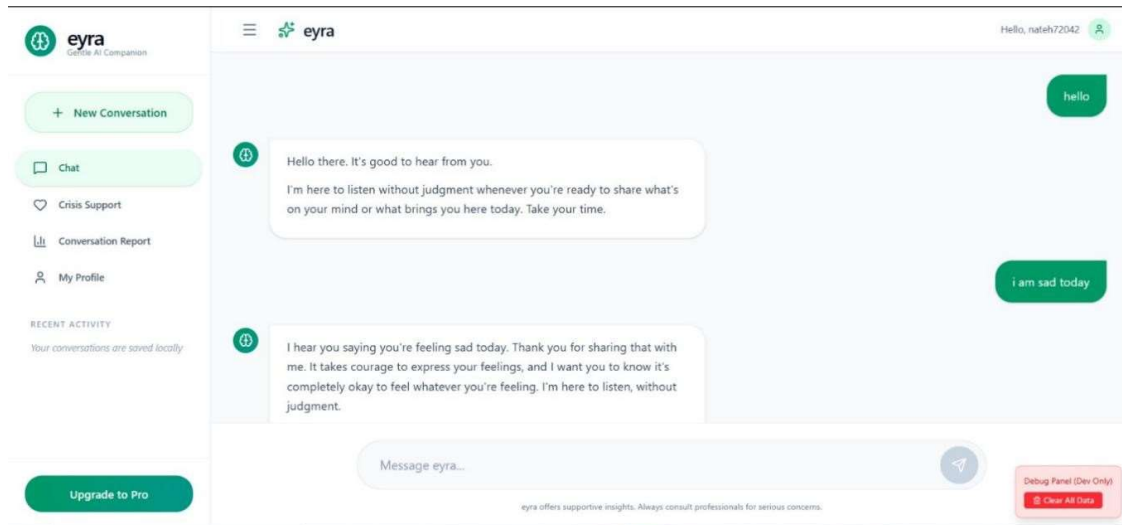


Figure 2: Illustrates the chat interface that allows users to interact with the platform and find assistance with emergency resources.



Figure 3: Uses multiple graphs to illustrate reports on mood trends, stress levels, and emotion distribution.

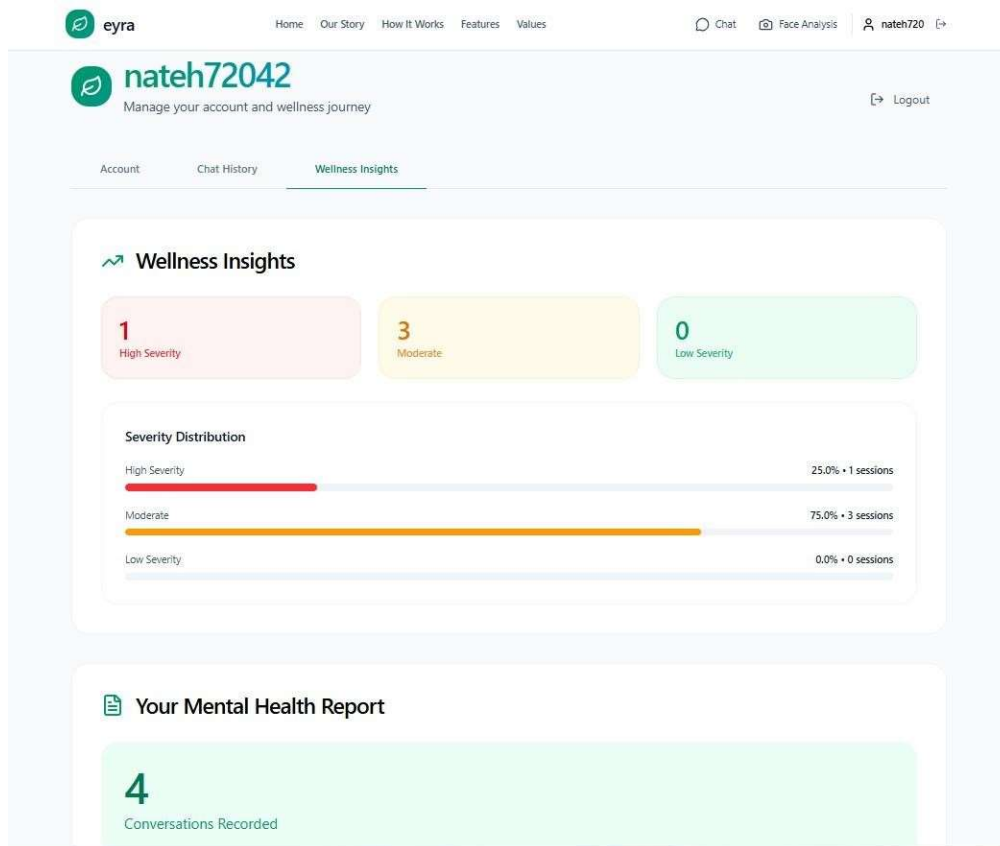


Figure 4: Displays total wellness data divided by levels of severity.

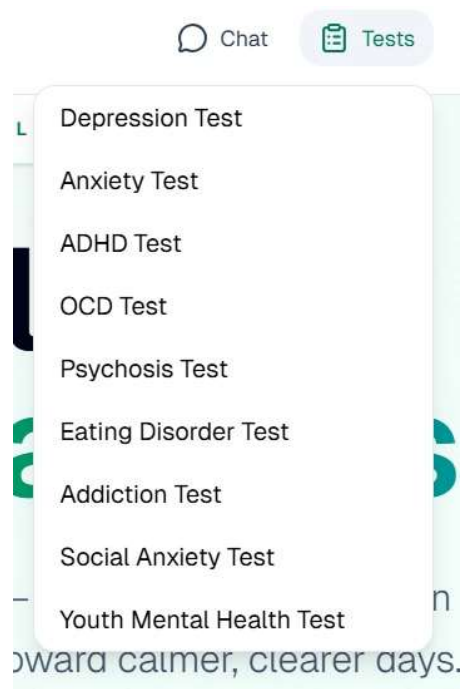


Figure 5: Shows different type of tests

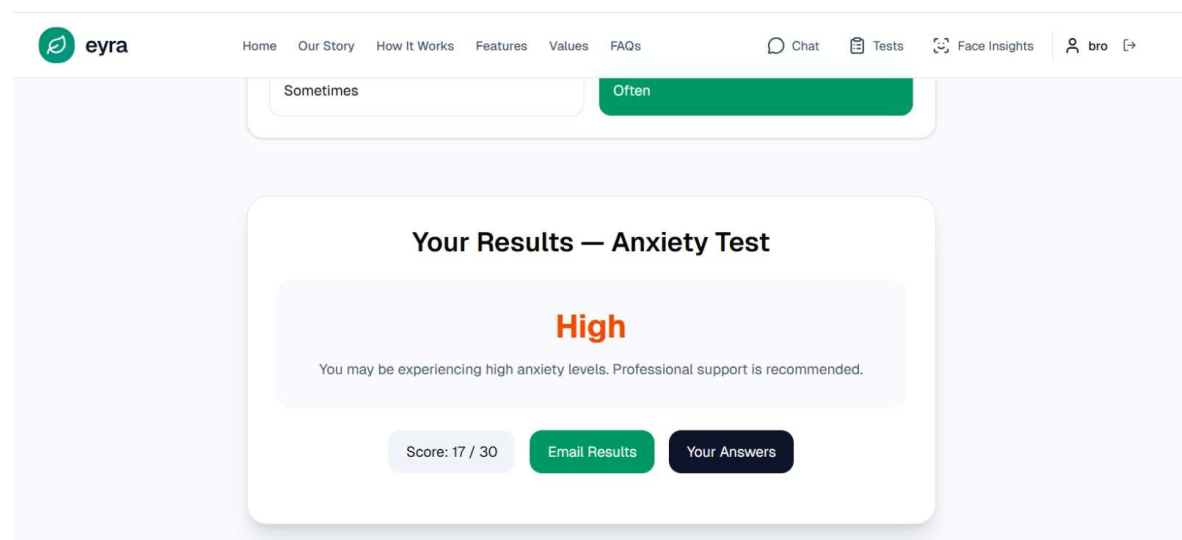


Figure 6: Shows results of tests

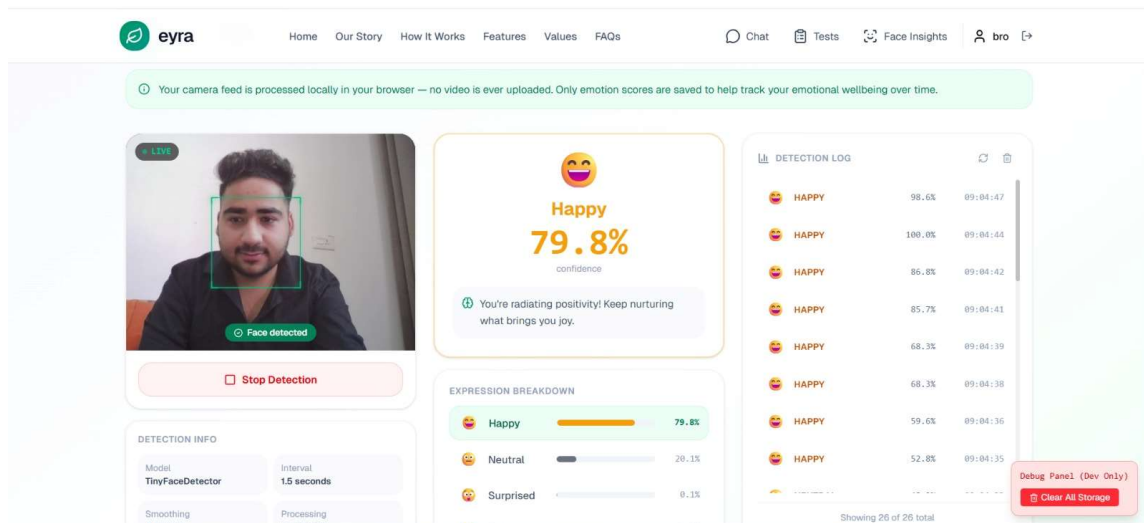


Figure 7: Illustrates the facial analysis feature that identifies emotions and gives feedback.

1.5 Technical/Methodology used

The application was built on the MERN software stack (MongoDB - database for storing information; Node.js - back-end processing; Express - back-end framework; React - presenting information/user interface). The application was developed using the MERN stack with MongoDB as the storage database, Node.js and Express to handle the backend processing, and React for presenting information (the user interface). The application also uses Google Gemini AI for the generation of responses & emotion analysis. Security will be achieved by using JWTs to authorize access to resources & encryption to protect user data.

1.6 Expected Outcomes & Benefits

This project is intended to give users an alternative to being able to effectively manage their emotions both in their everyday lives & in efforts to improve their situations by giving them the tools needed to effectively manage their life. It encourages open conversations about mental health without fear of judgment. While it does not replace professional therapy, it can act as an early support system. It also helps users identify emotional patterns and take necessary steps when required, making mental health support more accessible.

1.7 Original Contribution Statement

This project combines chatbot interaction, emotion detection, and explainable AI techniques to create a simple and practical system for supporting mental health in a responsible way.

1.8 Impact on Society

The mental health issues will be destigmatised due to the new system. Through dialogues that simplify the subject and create a comfortable space for people to share their experiences, it will prevent numerous risks. The new system, which raises awareness about mental problems, can improve overall health, particularly with youth and those that use technology frequently. The new system will create a more supportive and

easy-to-use environment to discuss mental health issues and it will enhance self-awareness and facilitate timely interventions.

1.9 Business Model and Market Potential

The marketplace has great potential because there are many people who would like to use a service focused on mental well-being, and this service is attractively priced and can reach a large number of customers. Thus, there is a high market potential for this software as a service opportunity.

Additionally, this service can be sold through several different platforms; therefore, the customer base will continue to grow as more individuals seek out low-cost digital mental health services within the mental health technology vertical.